CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

April/May 2012

EE - 302 Electrical Power System -II

Date: 03/05/2012, Thursday **Time: 10 a.m. to 1 p.m** **Maximum Marks: 70**

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

Q-1 What is sag in O/H lines? Discuss the disadvantages of providing too small or too large sag on line. Also discuss the effect of ice loading and temperature on sag of line. 7

Q-2 A. Justify the choice of bundled conductor as a member of transmission line. 5

B. A certain 3-phase equilateral transmission line has a total corona loss of 53 KW at 106 KV and a loss of 98 KW at 110.9 KV. What is the disruptive critical voltage between lines? What is the corona loss at 113 KV? 6

C. List out the advantages of suspension insulator over pin insulator 3

OR

Q-2 A. Write technical note on " Insulation Coordination" 7

B. Explain the term "Critical corona voltage" with reference to corona. 3

C. A 3-phase transmission line has conductors of radius 0.8 cm, spaced equilaterally 2.5 cm apart. If the dielectric strength of air is 30 KV/cm (Peak). Determine the critical corona voltage. Take relative air density factor $\delta=0.96$ and irregularity factor $m_0=0.94$ 4

Q-3 A. Explain the significance of tower footing resistance 3

B. List out the main specification of transmission line 4

C. Write technical note on " Radio Interference" 7

OR

Q-3 A. Explain the considerations for selection of type, size & location of generating station 6

B. List out the advantages of interconnection. 2

C. Draw any two well labelled bus bar arrangement 6

SECTION – II

Q -4	Write a Technical Note on Voltage Regulation and Explain any four methods of voltage regulation used on distribution systems in detail.	7
Q -5 (A)	Explain types of Distribution System Arrangements & also draw neat well labelled diagram of these systems.	7
(B)	A 2-wire d.c. distributor cable AB is 2 km long and supplies loads of 100A, 150A, 200A and 50A situated 500 m, 1000 m, 1600 m and 2000 m from the feeding point A. Each conductor has a resistance of 0·01 Ω per 1000 m. Calculate the p.d. at each load point if a p.d. of 300 V is maintained at point A.	7
	OR	
Q -5 (A)	A 2-wire d.c. street mains AB, 600 m long is fed from both ends at 220 V. Loads of 20 A, 40 A, 50 A and 30 A are tapped at distances of 100m, 250m, 400m and 500 m from the end A respectively. If the area of X-section of distributor conductor is 1cm2, find the minimum consumer voltage. Take ρ = 1·7 × 10−6 Ω cm.	7
(B)	A 2-conductor cable 1 km long is required to supply a constant current of 200 A throughout the year. The cost of cable including installation is Rs. (20 a + 20) per metre where 'a' is the area of X-section of the conductor in cm2. The cost of energy is 5p per kWh and interest and depreciation charges amount to 10%. Calculate the most economical conductor size. Assume resistivity of conductor material to be 1·73 μ Ω cm.	7
Q -6 (A)	Write a Technical Note on protection of distribution system.	4
(B)	What do you mean by economics of distribution system?	2
(C)	Write a Technical Note on planning and design of town electrification schemes.	8
	OR	
Q -6 (A)	What is meant by power system reliability?	2
(B)	Explain the advantages and limitations of Single Wire Earth Return (SWER) in connection with remote rural electrification.	6
(C)	List out the common causes of unsatisfactory performance of rural distribution system and also suggest the remedial measures for the same.	6

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